

Week 1 Worksheet

Math Review

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Exercise 1. Probability. Suppose you drop a match of length 1 cm onto a ruled sheet of paper whose lines are 1 cm apart. What is the probability that the match crosses a line on the paper?

Hints: First, figure out where the center of the match will land. Then, determine the angle of the match relative to the lines on the paper.

Exercise 2. Linear Algebra. Suppose V is a (complex) vector space. Let V^* be the space of all *linear* maps $V \rightarrow \mathbb{C}$. A map $\varphi : V \rightarrow \mathbb{C}$ is called **linear** if

$$\varphi(a\psi_1 + b\psi_2) = a\varphi(\psi_1) + b\varphi(\psi_2)$$

for any ψ_i in V and any complex numbers a and b .

- a) If V has dimension $n < \infty$, show that V^* is also a vector space and has dimension n .

Hint: Use a basis for V to construct a basis for V^* .

- b) Recall that if V is a vector space, a (complex-valued) **inner product** on V assigns to any ordered pair of two vectors ψ_1, ψ_2 in V a complex number $\langle \psi_1 | \psi_2 \rangle = \langle \psi_2 | \psi_1 \rangle^*$ such that if $\psi \neq 0$, then $\langle \psi | \psi \rangle > 0$. Show that in the situation of (a), an inner product on V is the same thing as a *bilinear* map, i.e. a map which is linear in both inputs, $V^* \times V \rightarrow \mathbb{C}$ such that its value is 0 if and only if one of the inputs is 0 (we say it's **nondegenerate**).

- c) Let $L^2(\mathbb{R})$ be the vector space of complex-valued, square-normalizable functions on the real line, $\psi : \mathbb{R} \rightarrow \mathbb{C}$. A function $\psi : \mathbb{R} \rightarrow \mathbb{C}$ is **square-normalizable** if

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx < \infty.$$

Explain how $L^2(\mathbb{R})$ is a vector space.

- d) For the vector space $V = L^2(\mathbb{R})$ of (c), show that there are functions in V^* which are not in V . This means V^* is *strictly larger* than V .
- e) What would go wrong if you tried to apply your argument for (a) to the vector space $L^2(\mathbb{R})$?

f) Check that the assignment

$$\langle \psi_1 | \psi_2 \rangle = \int \psi_1^* \psi_2 dx$$

is an inner product on $L^2(\mathbb{R})$.

g) Is there a bilinear map $L^2(\mathbb{R})^* \times L^2(\mathbb{R}) \rightarrow \mathbb{C}$ which recovers the inner product of (f)?